

UNIVERSITY OF CLAUDE · COURSE SYLLABUS

CSCE 671 — Human-Computer Interaction

Designing for the person who will use it, who is not you

AT A GLANCE

Credits: 3 credit-equivalents · **Modules:** 13 · **Term:** Semester 11 (with CSCE 679 and CSCE 632)

Prerequisites: CSCE 606 Software Engineering for the process material and CSCE 633 Machine Learning for the experimental design in Module 08; CSCE 679 and CSCE 632 run in parallel and extend Modules 02 and 05 respectively

Effort: 12–14 hours per week · 13 modules + 2 project weeks

Deliverable: A redesign of an interface you did not design, driven by observed failures rather than by opinion — with users watched before and after, the specific failures documented, and an honest account of what your evaluation cannot establish

Course description

Interfaces fail for reasons that are predictable in advance and invisible to their designers. **Module 01** establishes why: the designer knows the system's model and the user has only the interface, so every assumption the designer cannot see is one the user has to guess — and the resulting failures look like user error from the inside and like bad design from outside.

The first third is about what people are actually like. **Perception, attention, memory, and error are measured properties with hard limits** (Modules 02 and 03) — and a design that requires a person to notice something, remember something, or be careful will fail at a rate you can predict, which turns a great deal of design argument into arithmetic.

The second third is method. **Prototyping, usability evaluation, controlled experiments, and qualitative observation** (Modules 06 through 09) — and the governing fact is that your opinion about your own interface is worth very little, because you cannot un-know how it works. Watching five people use it is worth more than any amount of reasoning about it.

The third theme is that this course's subject includes the interfaces you build for other programmers. **Module 10** treats APIs, error messages, configuration, and tooling as user interfaces, because they are — and because that is the interface design most of this program's graduates will actually do.

And the closing position is about honest claims and about consequences. Module 12 covers deliberately harmful design — the techniques are the same ones, aimed differently — and **Module 13 is about what a usability study establishes**, because 'users liked it' is a statement about five people in a room.

Learning outcomes

On finishing this course you should be able to do the following without reference material.

- 1 Explain why designers cannot see their own interfaces' failures.
- 2 Apply what is known about perception to a layout.
- 3 Predict error from attention and memory limits.
- 4 Elicit and design against a user's mental model.
- 5 Apply the interaction design principles and say what each costs.
- 6 Prototype at the right fidelity for the question.
- 7 Run a usability test and analyse it.
- 8 Design a controlled experiment with human participants.
- 9 Use qualitative methods and know what they establish.
- 10 Design an API, an error message, and a configuration system.
- 11 Explain input and output modalities and their constraints.
- 12 Recognise and refuse manipulative design.
- 13 State what a usability finding honestly establishes.

Module schedule

One module per week for thirteen weeks, then two weeks for the final project. Each module is a slide deck, a set of lecture notes, mapped external lectures, and exercises that must run.

Wk	Module	Core question it answers
1	Why Interfaces Fail	Why can you not see the problems in your own interface?
2	Perception	Why did they not notice it?
3	Attention, Memory, and Error	What can you require a person to hold in mind?
4	Mental Models	What does the user think is happening?
5	Interaction Design Principles	Which principles are real, and when do they conflict?
6	Prototyping	What fidelity do you need?

Wk	Module	Core question it answers
7	Usability Testing	What happens when somebody actually tries it?
8	Controlled Experiments with People	Is design A actually better than design B?
9	Qualitative Methods	What is actually going on, and why?
10	Interfaces for Programmers	Why is this library hard to use?
11	Input and Output	Why is this interaction awkward on this device?
12	Manipulative Design	When does persuasion become manipulation?
13	Claiming Usability Honestly	You tested it. What do you know?

Projects

Project 1 — Watched, and diagnosed (after Module 6)

Watch people use an interface and diagnose the failures mechanically rather than by opinion.

Requirements

- **Five people observed** attempting the same three tasks, with the sessions recorded or carefully noted.
- **Every failure logged** with what the person expected, what happened, and what they did next.
- **Each failure attributed to a mechanism** from Modules 01 through 05 — **a missing signifier, an attention failure, a memory demand, a mental model mismatch** — not to 'confusing'.
- **A count by category**, so you know which mechanism accounts for most of the trouble.
- **Three things you predicted would fail and did not**, reported honestly.
- **And a redesign proposal per category**, with the mechanism it addresses named.

Done means

- **Failures attributed to mechanisms rather than described** — **which is the whole skill, and is graded hardest**. 'The button was confusing' is not a diagnosis.
- **Five people, not two**, and the sessions documented well enough that somebody else could count the same failures.
- **The wrong predictions reported**, because **a study in which you were right about everything was not a study**.
- **And the proposals tied to mechanisms**, so that each one is testable rather than a matter of taste.

Project 2 — Redesigned, and re-tested (after Module 12)

Implement the redesign and test whether it actually helped.

Requirements

- **The redesign built**, at whatever fidelity supports the test (Module 06 §2).
- **The same three tasks, with five new participants** — and **the same observation protocol**, so the comparison means something.
- **Task completion, time, and error counts** before and after, with the per-participant spread shown.
- **A qualitative account** of what changed in how people approached the tasks (Module 09).
- **The accessibility review** from CSCE 632's material, applied to your redesign.
- **And a limitation statement** in Module 13 §2's form.

Done means

- **New participants, not the same five**, because a second session with the same people measures learning rather than design.
- **The spread reported, not only the means** — **with five participants the mean is nearly uninformative** and the individual trajectories are the data.
- **Any regression reported: a redesign that fixed two things and broke a third is the normal outcome** and a complete project.
- **And the limitation statement specific: five people, one population, one set of tasks, in a room with you watching** — all four of which constrain the claim.

Assessment

Nobody grades this. The standard is therefore mechanical, and you apply it to yourself honestly or the credential is worth nothing.

Component	Weight	Standard
Module exercises	40%	Code runs and produces the specified output. Not 'I understand how I would do it.'
Projects	45%	In the portfolio repository, with a README showing results and a written note on what was hard.
Self-checks	15%	Answered in writing from memory. Recognition is not knowledge; writing exposes the difference.

Required materials

All free. Nothing in this course requires a purchase.

Resource	Role
Norman — The Design of Everyday Things , revised edition	The primary text for Modules 01, 04, and 05. The affordance, signifier, and conceptual model vocabulary this course uses comes from here, and the examples are still the best ones. Library copy.
Johnson — Designing with the Mind in Mind , 3rd edition	Modules 02 and 03. The perceptual and cognitive research, written for designers and honest about effect sizes — which is unusual in this area. Library copy.
Krug — Don't Make Me Think , revisited	Modules 05 and 07. Short, practical, and its discount usability testing chapter is the best introduction to Module 07 there is.
Nielsen Norman Group articles, and the heuristics (free)	Modules 05 and 07's working reference — the heuristics, the discount methods, and a large body of reported studies. Read the method sections, not only the conclusions.

Resource	Role
Lazar, Feng & Hochheiser — Research Methods in Human-Computer Interaction, 2nd edition	Modules 08 and 09. The methods treated properly, including the ethics and the statistics. Library copy.
The CHI proceedings (free via the ACM OA programme)	The field's primary literature. Module 13's exercises use it; read the limitations sections, which is where the honest claims are.

Tooling

- **Paper and a pen, first. Module 06 §1 argues that low-fidelity prototyping gets more and better feedback than a polished mock-up,** and the exercises require it before anything digital.
- **A screen recorder and a way to take notes while watching. Module 07's testing needs almost no tooling** — which is the point, and is why the barrier to doing it is not equipment.
- **Any prototyping tool you already know** for the higher-fidelity work. **The tool does not matter; the fidelity decision does** (Module 06 §2).
- **Python with `scipy.stats` and `statsmodels`** for Module 08. **You will be analysing very small samples,** which is where the assumptions matter most.
- **Three to five people who will let you watch them use something. This is the real prerequisite,** and Module 07 §2 explains why five is the number.
- **And an interface you did not design, that you can change. Project 2 requires one** — your own code's interface is acceptable only if somebody else can be watched using it.

Academic honesty, restated for a solo student

There is no one to cheat but yourself, which makes it easier, not harder. Two rules: copy code only after you have written a worse version yourself, and never let a working result stand in for an understood one. When you paste something you do not understand, write a TODO naming what you do not understand, and go back to it.